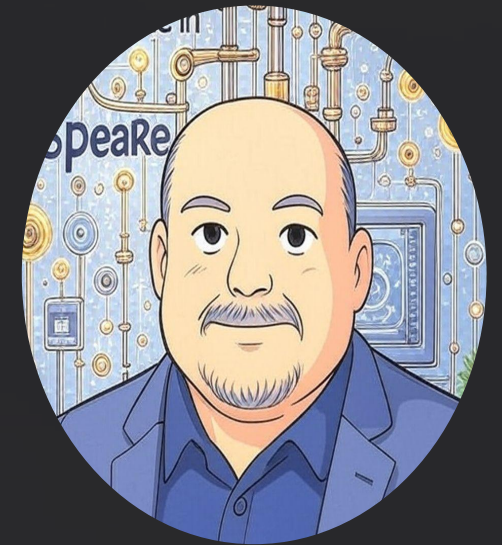


# Aaron Meis

- Tech Evangelist
- Not a physicist.
- Interested in applications of quantum encryption, communication, and data manipulation.





# Quantum Computing: Bridging Today's Reality with **Tomorrow's Potential**

Aaron Meis | Associate Director, DT AI Team, Collins Aerospace

April 8, 2025



# Overview

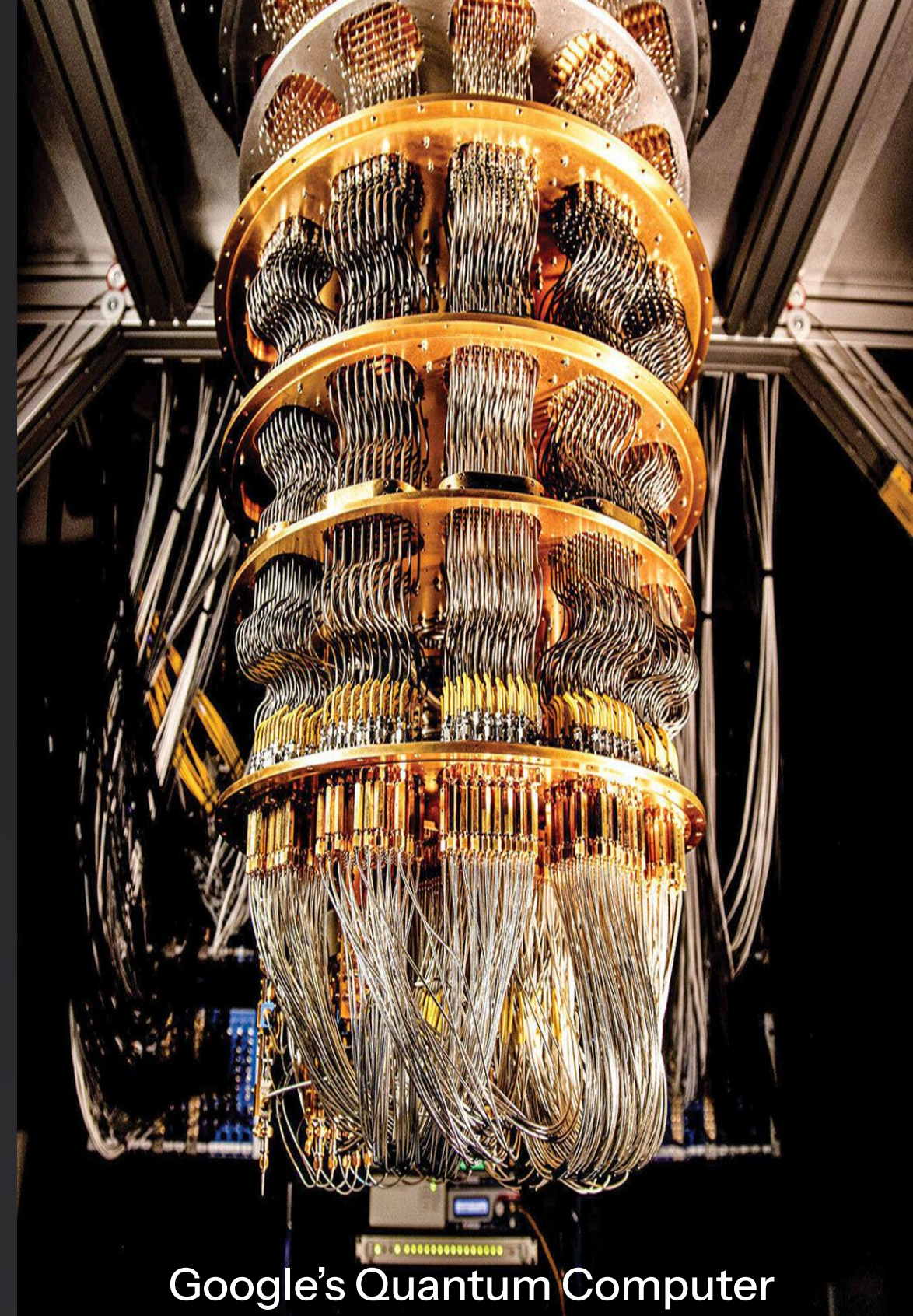
1. What is quantum computing?
2. What challenges does quantum computing solve or awaken?
3. Why is quantum important for Iowa?
4. What do we need to do now to prepare for quantum?

# What is quantum computing?

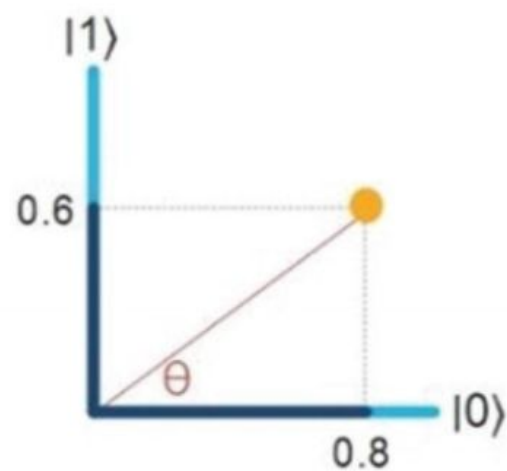
Quantum computers solve problems too computationally complex for our existing machines.

## Business Applications

- Secure Communication
- Enhanced cybersecurity
- Optimization problems
- Improved risk modeling
- Quantum Enhanced Machine Learning/AI

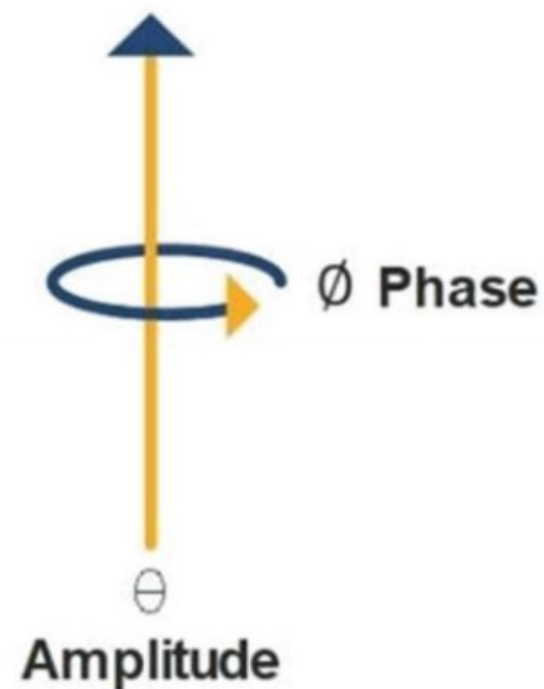
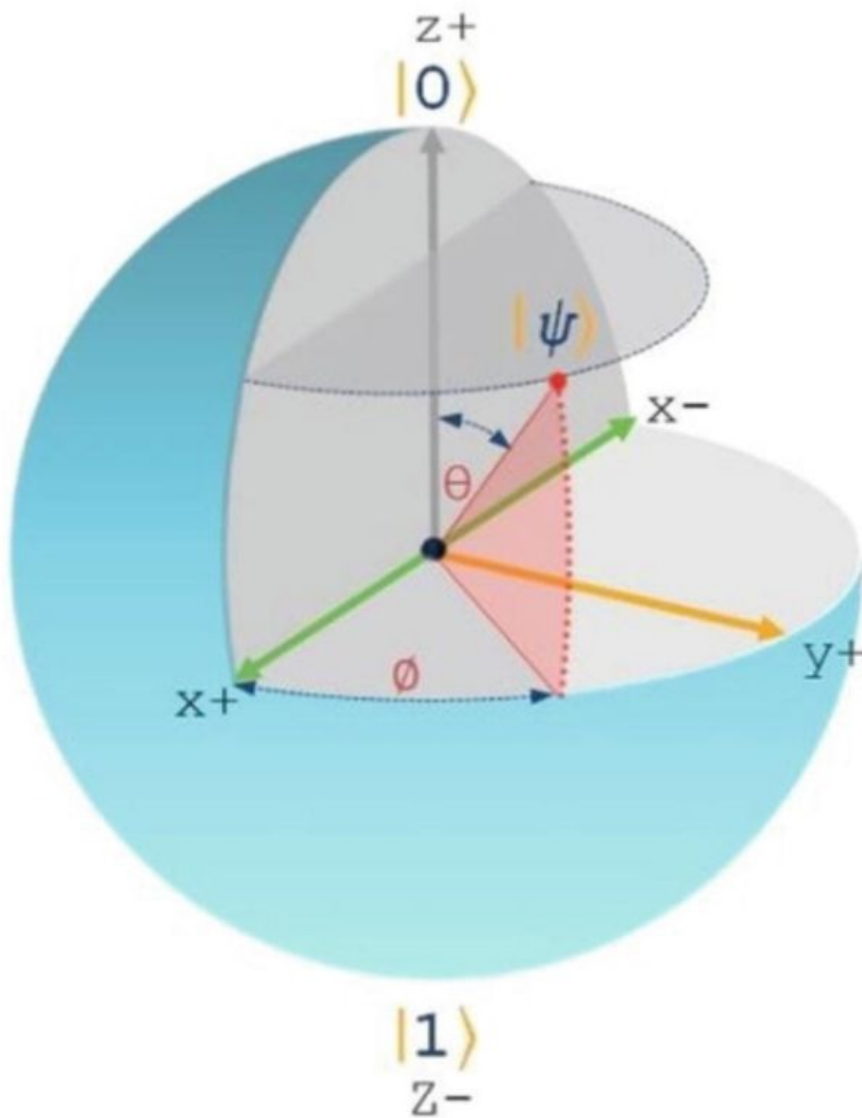


Google's Quantum Computer



$$|\psi\rangle = 0.8|0\rangle + 0.6|1\rangle$$

Probability Amplitudes (when squared)



The Bloch sphere is a visual representation of qubit states as points on the surface of a unit sphere.

**Figure 1-3** The Bloch sphere

# Classical vs. Quantum Computing:

## A Fundamental Shift

|                  | Classical Computing                 | Quantum Computing                                |
|------------------|-------------------------------------|--------------------------------------------------|
| Basic Unit       | Bit (0 or 1)                        | Qubit (superposition of 0 and 1)                 |
| Information      | Definite state                      | Probabilistic state                              |
| Computation      | Sequential                          | Parallel (due to superposition and entanglement) |
| Problem Solving  | Limited by computational complexity | Suited for specific complex problems             |
| Error Correction | Well-established methods            | Still under development                          |

PQC  
Ready!

### Important Distinctions:

- Not a replacement for classical computers
- Specialized for probabilistic problem-solving
- Best for simulating natural phenomena

### Hybrid Approach:

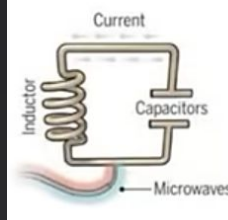
- Combine classical and quantum computing
- Dispatch tasks based on computational complexity

# The Engineering Question...

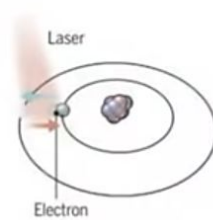
## Seven Major Quantum Computer Building Methods

- Varying characteristics
- Some require extreme cooling
- Some operate at room temperature
- Different scalability potential
- Most cloud solutions and open source tools are at the Big 3: IBM, Google, Microsoft

## Platforms for Quantum Computing Technology



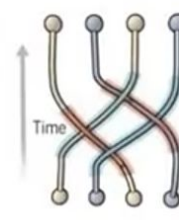
**Superconducting**  
A resistance-free current oscillates back and forth around a circuit loop. An injected microwave signal excites the current to superposition states.



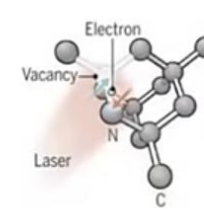
**Trapped ions**  
Electrically charged atoms or ions have quantum energies that depend on the location of electrons. Tunes laser cool and trap the ions and put them in superposition states.



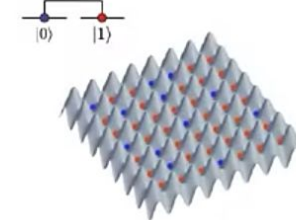
**Silicon quantum dots**  
These "artificial atoms" are made by adding an electron to a small piece of pure silicon. Microwaves control the electron's quantum state.



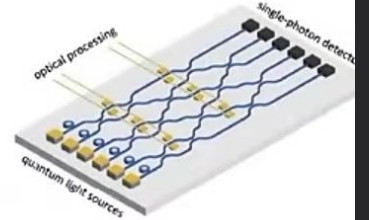
**Topological qubits**  
Quasiparticles can be seen in the behavior of electrons charged through semiconductor structures. Their braided paths can encode quantum information.



**Diamond vacancies**  
A nitrogen atom and a vacancy add an electron to a diamond lattice. Its quantum spin state along with those of nearby carbon nuclei can be controlled with light.



**Neutral atoms**  
Neutral atoms store qubits within electronic states. Interactions through excitation to Rydberg states.



**Photonics**  
Photonic qubits interact via linear elements

IBM Q  
Google  
rigetti

QUANTINUUM  
IONQ

intel Microsoft

QDTI  
Quantum Diamond Technologies, Inc.

IQuEra  
Computing Inc.

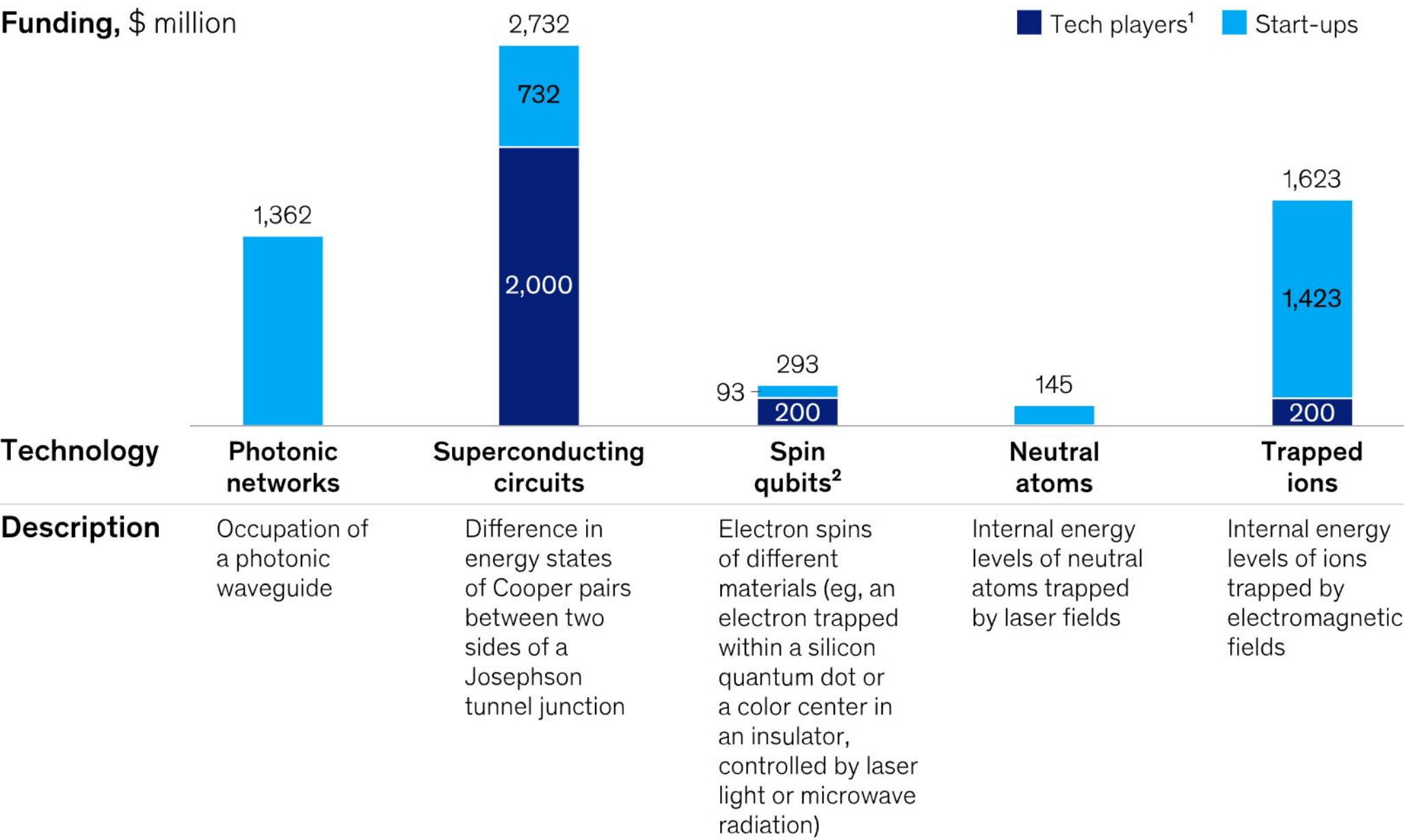
PASQAL

Ψ PsiQuantum

XANADU



# Five main qubit technologies are competing to build a scalable universal quantum computer.

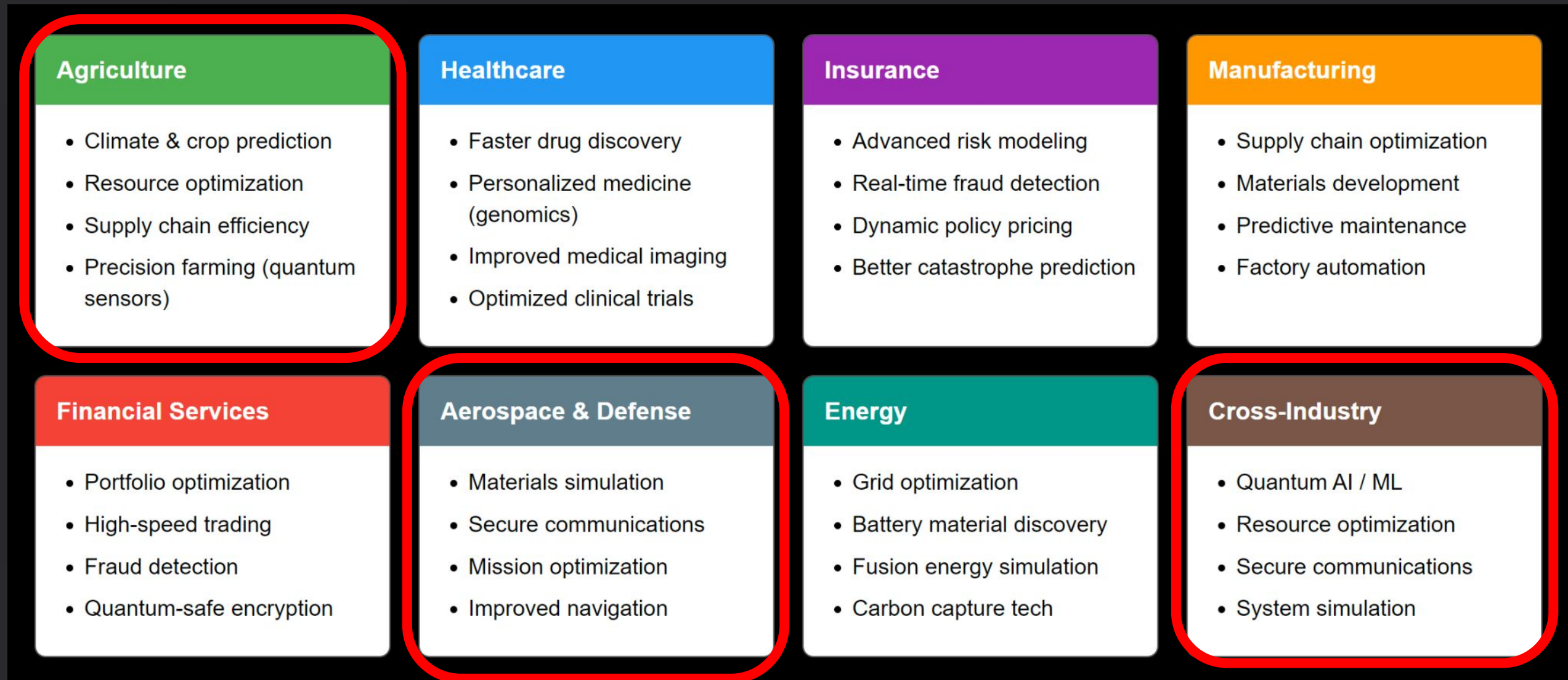


<sup>1</sup>Assumptions: \$500 million per major player (Google, IBM, Alibaba, AWS) and \$200 million per medium player (Honeywell before merger with CQC into Quantinuum, Intel).

<sup>2</sup>We examine electron spins in silicon quantum dots because other spin qubits are generally not considered for applications in quantum computing. For example, nitrogen-vacancy (NV) centers in diamond are unlikely to be a good qubit for computing; however, they can serve as quantum sensors.

Source: Capital IQ; Crunchbase; Pitchbook; McKinsey analysis

# Quantum Technology for Iowa's Economy



***“Our clients will get the value, whether it's material discovery or better batteries or better fertilizers or better drugs, that value will be accrued by our clients.” Arvind Krishna, IBM CEO (Mar 2, 2025)***

# Quantum Computing:

## Predicting Our 2030 Headlines

### Quantum Tech Helps Iowa Farmer Boost Yields by 30%

WITH QUANTUM COMPUTING, IOWA FARM OPTIMIZES WATER AND FERTILIZER USE—SETTING A NEW STANDARD FOR SUSTAINABLE AGRICULTURE IN THE MIDWEST.

### Iowa Aerospace Firm Uses Quantum Simulations to Extend Aircraft Range

BREAKTHROUGH ALLOY AND SECURE QUANTUM COMMUNICATIONS PUT THE STATE ON THE MAP AS A DEFENSE INNOVATION LEADER.

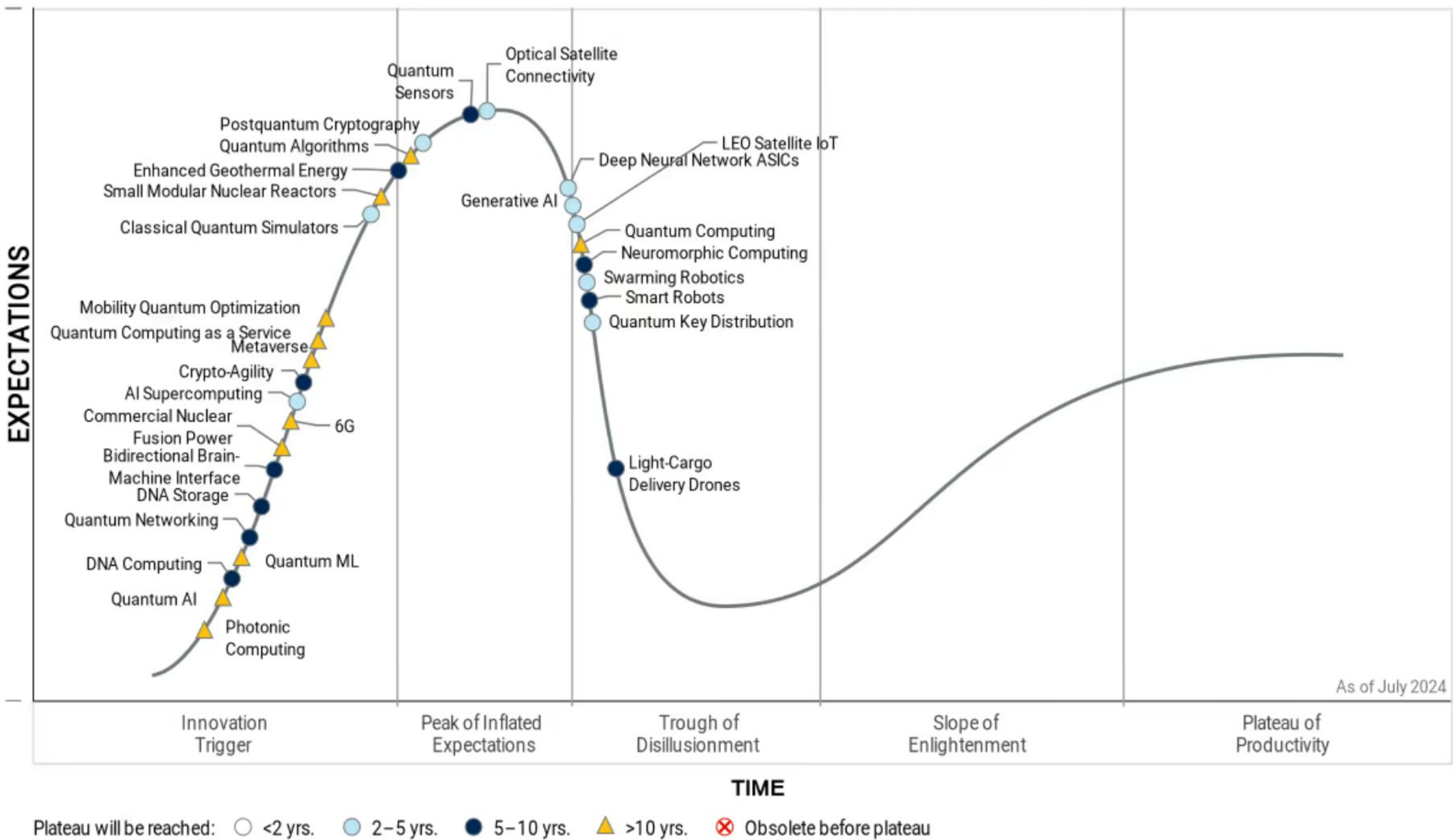
### Quantum AI Hub in Des Moines Cuts Energy Use Across Industries

NEW RESEARCH CENTER HELPS IOWA SLASH ENERGY CONSUMPTION BY 15% WHILE IMPROVING EFFICIENCY IN AGRICULTURE, MANUFACTURING, AND HEALTHCARE.



# Current state of Quantum

## Hype Cycle for Deep Technologies, 2024





With current supercomputers, breaking an advanced encryption would take **billions of years.**



With sufficiently capable quantum computers, this time is **hours or days.**

# Old Locks Break, Quantum Shields



## Current Encryption Vulnerability

- "Harvest now, decrypt later" threat
- 100% of banking systems (RSA/ECC) at risk
- Crypto wallets (ECDSA) exposed on blockchain
- All HTTPS traffic vulnerable



## Quantum-Proof Locks

- Post-Quantum Cryptography (PQC) protects data today
- NIST published quantum-proof standards, replacing vulnerable mathematical problems
- Implementation can begin now

As organizations begin the transition to post-quantum cryptography over the next year, **agility will be crucial to ensure systems are prepared for continued transformation**, particularly as the NIST continues to expand its toolbox of post-quantum cryptography standards.”

*Ray Harishankar, IBM Quantum Fellow*

***Begin your quantum-ready security assessment today—implement PQC protocols before your data becomes vulnerable to tomorrow's threats.***

Quantum in **Iowa**  
and nearby.



## Toor receives \$1M EPSCoR award to improve quantum computing

The field of quantum information science and engineering (QISE) is seen as critical for the future.

Monday, September 9, 2024

Fatima Toor, University of Iowa's Lowell G. Battershell Chair in Laser Engineering, has received a \$999,985 grant to bring next generation computing closer to reality.

Quantum computing allows the tackling of complex calculations – numerous possibilities at once – that are beyond the reach of traditional computer processors, such as those found in laptops and smartphones.

The field of quantum information science and engineering (QISE) is seen as critical for the future with a significant impact on U.S. scientific, economic, and national security infrastructures. Yet, gaps remain in making quantum computing a practical resource.

"The building blocks of quantum computers are qubits," said Toor, associate professor of electrical and computer engineering and faculty affiliate of the Iowa Technology Institute. "These qubits are typically based on superconducting materials that operate best at really low temperatures - liquid helium (He) temperatures - in a dilution refrigerator."



# Quantum Midwest Revolution. Iowa is **next.**

## NVIDIA Quantum Research Center

Integrating quantum hardware with AI supercomputers

## Abu Dhabi's Quantum Solver

17-qubit system solving large-scale optimization problems

## Poland's Quantum-Encrypted Connection

First domestic QKD technology implementation

## Startup to build massive quantum campus on Chicago's South Side



SciTechDaily

## A 56-Qubit Quantum Computer Just Did What No Supercomputer Can

Researchers have achieved a major quantum computing breakthrough: certified randomness, a process where a quantum computer generates truly...

3 days ago



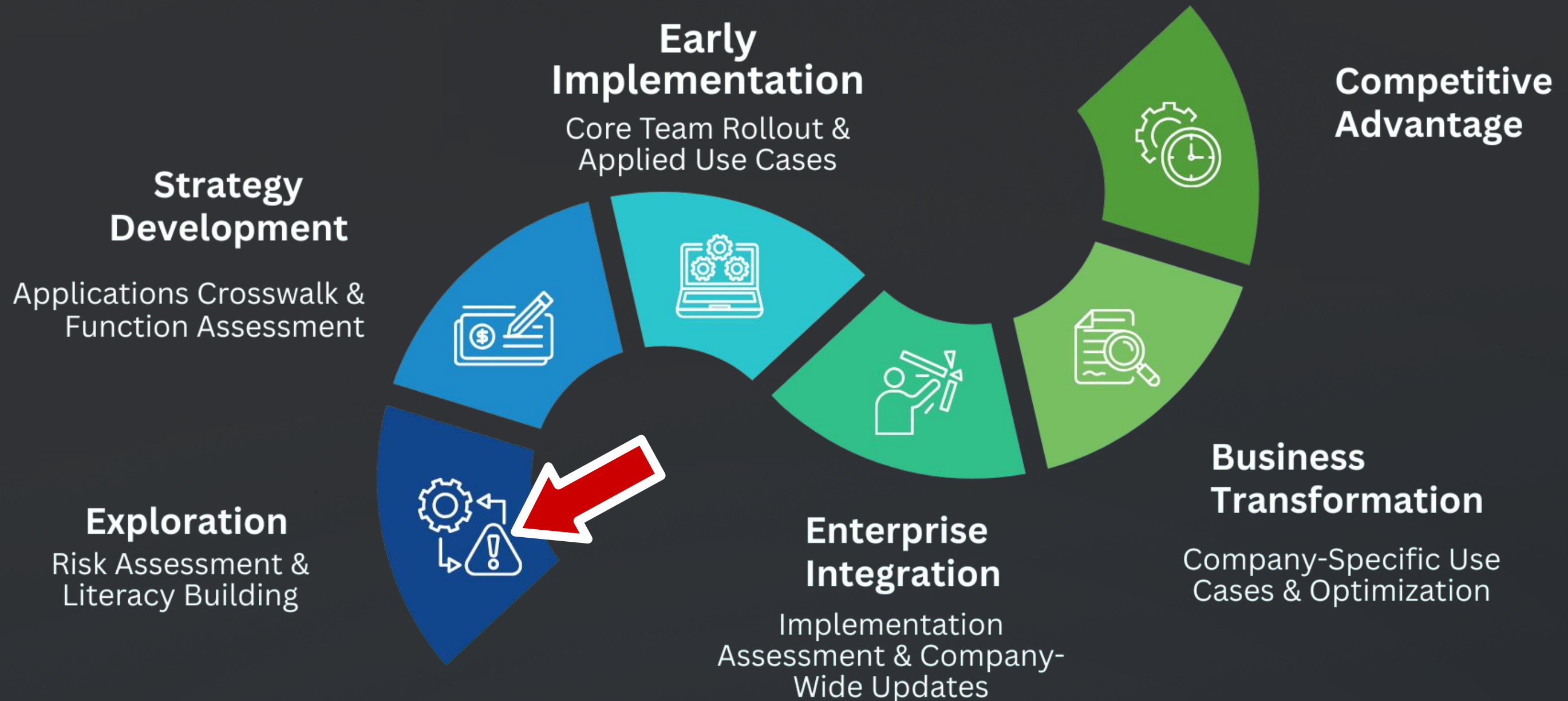
What do we need to do now  
to be **quantum ready?**

## DARPA Picks Firms for Quantum Computer Development Project

BY ARTHUR MCMILLEN • APRIL 4, 2025 • 3 MINS READ



# Emerging Tech Roadmap

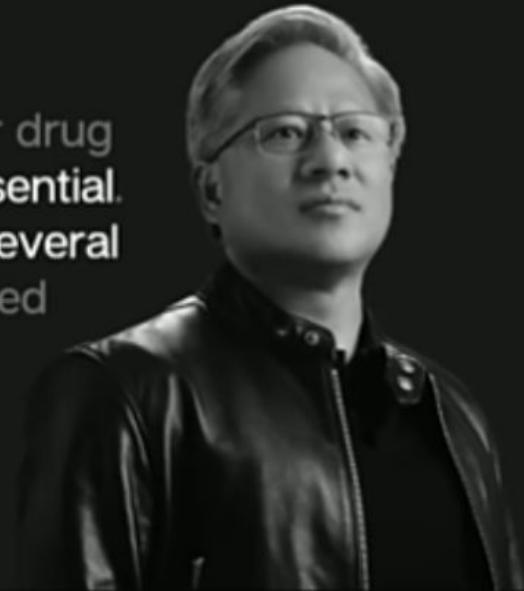


# How to explore and investigate Quantum today?

| Service Name                                                                                                                                                                                                                                                                                                 | Service Name                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>IBM Quantum Platform</b><br><br>Free tier available; Premium access via IBM Cloud billed per second (e.g., ~\$1.60/min for 127-qubit systems, ~\$96/hr). Exact rate depends on Quantum Allocation Units (QAUs).    | <br><b>Amazon Braket (AWS)</b><br><br>On-demand QPU pricing varies: e.g., IonQ Aria ~\$0.03/shot + \$0.30/task (~\$1,000-\$2,000/hr for heavy use); hourly reservation ~\$15-\$30/hr depending on QPU. |
| <br><b>Azure Quantum (Microsoft)</b><br><br>Pay-as-you-go: e.g., IonQ Aria ~\$25,000/month subscription (~\$34/hr amortized) + per-AQT rates (~\$1,000-\$2,000/hr for intensive tasks). Free \$500 credits for new users. | <br><b>Google Quantum Service</b><br><br>Limited access for approved users; no public hourly pricing. Estimated ~\$1,000-\$5,000/hr based on industry norms for exclusive use.                         |
| <br><b>D-Wave Leap</b><br><br>Starts at ~\$2,000/hr for QPU time; hybrid solver access via subscription (~\$10-\$20/hr amortized). Free 1-min trial included.                                                           | <br><b>IonQ Quantum Cloud</b><br><br>Via AWS/Azure: ~\$1,000-\$2,000/hr for on-demand use (e.g., 2,500 shots on Aria ~\$75.30/task). Direct access pricing unpublished, likely similar.              |

Quantum computing can solve exponential [...] problems, like [...] simulating atoms and molecules for drug discovery [...] **error correction is essential.** [...] to solve meaningful problems, **several million physical qubits** will be required

Jensen Huang,  
NVIDIA CEO



We know we can't even understand the basic structure of molecules today[...] I am confident one day quantum computing will drive progress there. [...] You would need to build a **fault-tolerant quantum computer** [...] to run more **complex algorithms**

Sundar Pichai,  
Alphabet CEO



Quantum computing] is going to take off like a rocket ship ... as we scale towards **a million qubits**, I think we've got some fundamental issues in **error correction**, control [...]

Arvind Krishna,  
IBM CEO

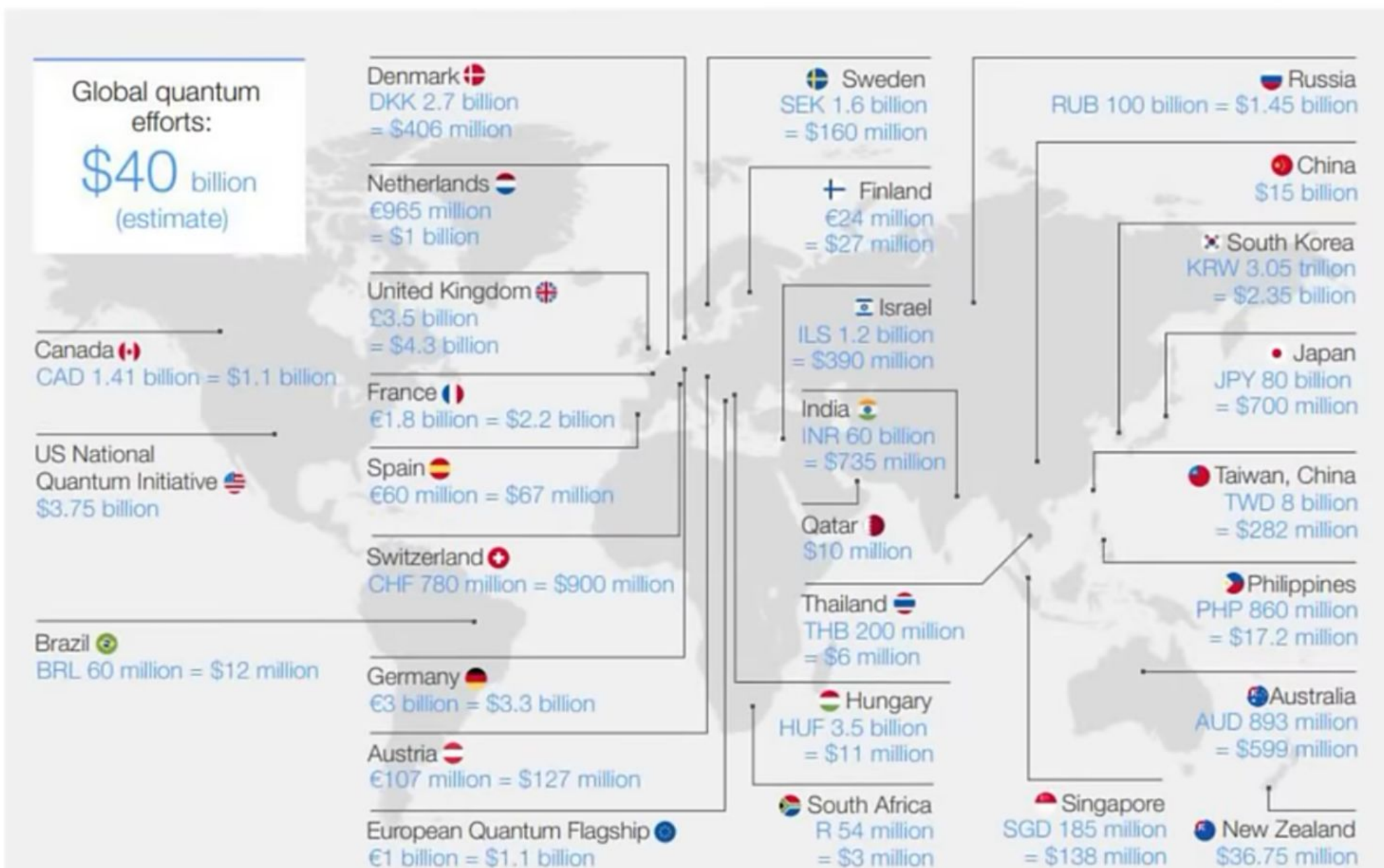


How do we efficiently capture carbon from the atmosphere [...] lossless power grids [...] Even if we combined all the world's computational power, these challenges would still take longer than the lifetime of the universe to solve. [...] **a quantum computer** [...] would be **so powerful** it's able to **model nature itself.**"

Satya Nadella,  
Microsoft CEO



# Global investment in Quantum Technology



**Note:** Not exhaustive; timelines for funding vary by country.

**Sources:** "Overview of Quantum Initiatives Worldwide 2023", QURECA, 19 July 2023, <https://qureca.com/overview-of-quantum-initiatives-worldwide-2023/>;  
Department of Industry, Science and Resources, Australia; ETH Domain (ETH Zurich, EPFL, PSI).



# Action Plan



## Prioritize PQC Implementation

Assess vulnerable systems. Develop migration plans for post-quantum cryptography.



## Explore Hybrid Solutions

Implement dual classical/quantum-resistant approaches. Ensure seamless transition.



## Develop Quantum Literacy

Train technical workforce. Educate business leaders on quantum opportunities.



## Foster Partnerships

Build collaborations with quantum research institutions. Establish public-private initiatives for technology transfer.



## Stay Informed

Monitor quantum computing advancements. Adjust strategy based on emerging standards and technologies. Follow standards published from NIST.

Prioritizing quantum R&D in land-grant universities and AgriTech incubators could position Iowa as a national leader. Invest in STEM education to create a quantum-ready workforce.



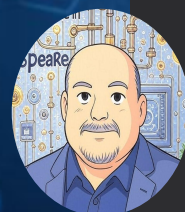
FAQ

# Thank you.

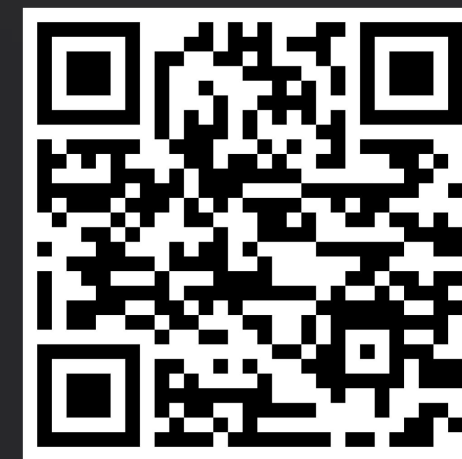
**“Prioritize Post-Quantum  
Cryptography (PQC)  
implementation”**

**Let's protect Iowa's future.**

Questions?



Aaron Meis



| Textbook algorithms          | Notebook                                                         | References                                                    |
|------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|
| Bell's Inequality            | <a href="#">Bells_Inequality.ipynb</a>                           | <a href="#">Bell1964</a> ,<br><a href="#">Greenberger1990</a> |
| Bernstein–Vazirani           | <a href="#">Bernstein_Vazirani_Algorithm.ipynb</a>               | <a href="#">Bernstein1997</a>                                 |
| CHSH Inequality              | <a href="#">CHSH_Inequality.ipynb</a>                            | <a href="#">Clauser1970</a>                                   |
| Deutsch-Jozsa                | <a href="#">Deutsch_Jozsa_Algorithm.ipynb</a>                    | <a href="#">Deutsch1992</a>                                   |
| Grover's Search              | <a href="#">Grovers_Search.ipynb</a>                             | <a href="#">Figgatt2017</a> , <a href="#">Baker2019</a>       |
| QAOA                         | <a href="#">Quantum_Approximate_Optimization_Algorithm.ipynb</a> | <a href="#">Farhi2014</a>                                     |
| Quantum Circuit Born Machine | <a href="#">Quantum_Circuit_Born_Machine.ipynb</a>               | <a href="#">Benedetti2019</a> , <a href="#">Liu2018</a>       |
| QFT                          | <a href="#">Quantum_Fourier_Transform.ipynb</a>                  | <a href="#">Coppersmith2002</a>                               |
| QPE                          | <a href="#">Quantum_Phase_Estimation_Algorithm.ipynb</a>         | <a href="#">Kitaev1995</a>                                    |
| Quantum Walk                 | <a href="#">Quantum_Walk.ipynb</a>                               | <a href="#">Childs2002</a>                                    |
| Shor's                       | <a href="#">Shors_Algorithm.ipynb</a>                            | <a href="#">Shor1998</a>                                      |
| Simon's                      | <a href="#">Simons_Algorithm.ipynb</a>                           | <a href="#">Simon1997</a>                                     |

| Advanced algorithms | Notebook                                                    | References                                              |
|---------------------|-------------------------------------------------------------|---------------------------------------------------------|
| Quantum PCA         | <a href="#">Quantum_Principal_Component_Analysis.ipynb</a>  | <a href="#">He2022</a>                                  |
| QMC                 | <a href="#">Quantum_Computing_Quantum_Monte_Carlo.ipynb</a> | <a href="#">Motta2018</a> , <a href="#">Peruzzo2014</a> |

| Auxiliary functions      | Notebook                             |
|--------------------------|--------------------------------------|
| Random circuit generator | <a href="#">Random_Circuit.ipynb</a> |